FPGA Implementation of a Signal Synthesizer for Driving a High-power Electrostatic Motor

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***Abstract*—This paper describes a motor driver circuit for a high-power electrostatic motor using Field Programmable Gate Array (FPGA). Electrostatic actuators are popular in the field of micron or nano technologies but some high-power electrostatic motors have been developed for normal-sized applications. *Dual Excitation Multiphase Electrostatic Drive (DEMED)* is one of the strongest actuators among them. There are various driving methods for the motor (DEMED), however, regardless of driving methods, developing driving circuits is one of the difficulties for practical applications. To operate the motor, one or two three-phase signals are required and amplitudes of them must be a few kV. Hence multi-channel signal synthesizers and high-power amplifiers are needed for driver circuits. In addition, portability is required for some applications as well. In this research, therefore, quad Direct Digital Synthesizers (DDS) are built inside the one FPGA. In the performance evaluation, this driver circuit generates motor driving signals based on the position command that is given by a higher-level controller. It is confirmed that the electrostatic motor operates following reference position data, by using this FPGA-based motor driver.**

I. Introduction

Electrostatic actuators are widely used in the field of Micro Electro Mechanical Systems (MEMS) because the electrostatic force has some advantages in that field compared with the electromagnetic force [1–9]. On the other hand, it is considered that the electrostatic force is generally unfitted to actuate normal-sized equipments. However, some groups have developed electrostatic actuators which can be applied to normal-sized equipments by accumulating micro electrostatic actuator elements [10–12]. Among them, *dual excitation multiphase electrostatic drive (DEMED)* proposed by Niino et al. [10] is one of the strongest electrostatic actuators. It consists of a pair of thin flexible films, on which three-phase electrodes, aligned in a micro-scale pitch, are accumulated. This motor can generate sufficient thrust force or torque to actuate normal mechatronic equipments. Hence this motor is called high-power electrostatic motor. In addition, the motor contains no ferromagnetic materials, so it is expected to be used for special applications in strong or precise magnetic fields [13], [14], e.g. a motion generator in an MR environment.

One of the difficulties in developing a practical system using the electrostatic motor is the driver circuit because the motor requires a few kV ac voltages. Some driver circuits have been developed that can be categorized into three groups based on operation methods; *single frequency operation*, *modulated single frequency operation*, and *dual frequency operation*.

Single frequency operation is the most simple method. A single three-phase high-voltage signal is applied to both electrodes of the two films and then films move relatively to each other. A relative velocity of the films is proportional to the applied signal frequency. One of disadvantages of this method is that it requires high-voltage amplifiers that have frequency range starting from DC. Such an amplifier costs a lot and the size is huge compared with the motor. Another problem is that the motor performance becomes unstable at a low speed.

Modulated single frequency operation is an extended method of the previous method. An advantage of this principle is that small-sized transformers can be used to amplify signals, instead of the high-voltage amplifiers, that leads to reduction of cost and size of the driving circuits. In addition, the problem of the unstable performance at a low speed is solved. But there is one problem in its performance; a generated thrust force is reduced by half of normal *single frequency operation*.

The third principle is *dual frequency operation*. In this principle, small-sized transformers can be used as amplifiers, as well as *modulated single frequency operation*. In addition, both performance problems described above do not happen in this method, so this method shows the best performance among the three methods. However, it requires two three-phase high-voltage signals that makes the driving system complex; double amount of signal generators and amplifiers are needed.

Some studies have been done to solve this problem regarding complexity of *dual frequency operation*. Yamamoto et al. [15] developed a motor driver circuit using a modified analog Phase-Locked Loop (PLL) circuit and transformers. But the analog based signal generator controlled by PLL is less flexible in its configuration, and therefore, it cannot cover a wide range of the motor operation. Nishijima et al. [16] and Zhang et al. [17] developed a digital driver circuit using Direct Digital Synthesizers (DDS). DDS has more flexibility than analog based signal generators. It is easier to generate signals with wide range of frequencies. To drive the electrostatic motor in dual frequency operation, six DDSs are required and they have to be operated in synchronization to generate two three-phase signals. In previous developments, six commercially available DDS chips are controlled by micro controllers. In such configurations, communication speed between a micro controller and DDS chips restricts control rates of DDSs and,

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in turn, the motor control rate.

In this paper, multi DDSs are implemented on the FPGA (Field Programmable Gate Array) to solve the above problem. In the next section, the paper reviews driving methods of the motor and previous developments of driver circuits. In section III, the new driver circuit using FPGA is described. Then the evaluation result of the performance is shown in section IV.

II. Structure and Driving Methods of the Electrostatic Motor

In this section, the basic structure and the three driving methods of the electrostatic motor is described from a viewpoint of driver circuit. Configurations of driver circuits differ according to the driving method used.

A. Motor structure

The motor consists of a pair of electrode films which have the same structure. On the films, three-phase electrodes are aligned in the longer direction of the film as shown in Fig. 1. The pitch of three-phase electrode generally ranges from tens to hundreds of microns.

The motor is driven by applying high-voltage three-phase signals to the three-phase electrodes. Depending on the signals used, driving methods can be classified into three types. Regardless of methods, one to two kV three-phase high-voltage signals are required to obtain practical thrust force.

B. Single frequency operation

The simplest principle is *single frequency operation* as shown in Fig. 1 [10]. Three-phase voltages are applied to the electrodes of both films but they are connected in reversed orders to two films. Application of a three-phase voltage to a three-phase electrode generates traveling electrostatic potential wave. In this operation method, traveling potential waves are excited on both films, but their traveling directions are opposite due to the reversed connections. Electrostatic interaction between two potential waves generate tangential force that tries to keep relative position of the two waves. As a result, the slider will be actuated by the electrostatic interaction so that the relative velocity between the two potential waves becomes zero. Since the traveling velocity of the potential waves are $\pm 3pf$, the resultant slider velocity is $6pf$, in which p is the electrodes pitch and f is the frequency of applied three-phase signals. This indicates that the motor is a synchronous motor; the velocity of the motor is proportional to the frequency of the applied three-phase signal.

In this principle, the driver circuit needs one multi-phase signal generator and three high-voltage amplifiers, which is the simplest among the three operation methods. However, since frequencies of the applied signals ranges from DC to 1 kHz because the velocity of the motor is proportional to the applied signal frequencies, the high-voltage amplifiers need to have frequency bandwidth starting from DC. Such high voltage amplifiers are commercially available, but they are typically quite expensive and bulky. Another solution to obtain high voltage signals is to utilize a combination of low-voltage amplifiers and transformers. But the requirement of the bandwidth makes it almost impossible to utilize transformers since transformers core can easily saturate at low frequency ranges.

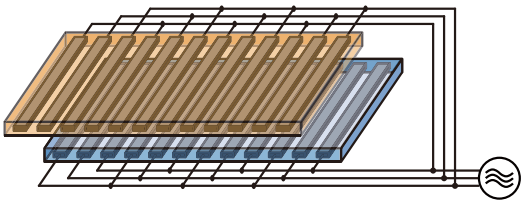


Fig. 1. Connection diagram of *Single frequency operation* [10]

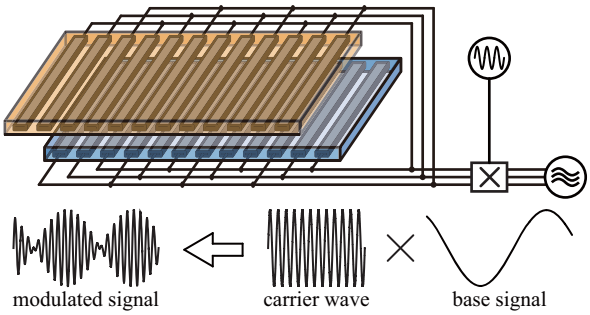


Fig. 2. Connection diagram of *Modulated Single frequency operation* [18]

Another problem reported in a literature is that the motor driving performance becomes unstable if it is driven at a very low frequency such as 0.1 Hz or smaller (that is about 0.1 mm/s or slower for driving velocity). This would be because of a slight conductivity of the insulating films. This means that single frequency operation has a lower limit in its operation velocity.

C. Modulated single frequency operation

To solve above problems, an extended principle of *single frequency operation* was proposed, that is *modulated single frequency operation*. In this method, the three-phase signal of the single frequency operation is modulated by a high-frequency carrier wave, as shown in Fig. 2 [18]. The motor operation is determined by the envelop curves of the modulated wave. Therefore, if we focus on the envelope curves, the operation principle is complete same as the single frequency operation. However, the modulated signals have much higher frequency than the single frequency operation, the signals can be easily amplified by small transformers. In addition, the unstable performance at low speeds is also cleared.

However, in this operation method, the motor generates just a half of the thrust force compared with *Single frequency operation*, since the effective value of the signal becomes $1/\sqrt{2}$ by the modulation (and the thrust force is proportional to the square of the applied voltage).

D. Dual frequency operation

The third operation method (that was actually proposed in the first) is *Modulated single frequency operation*, in which

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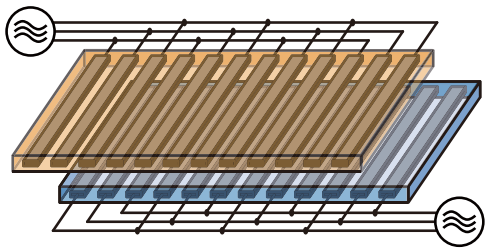


Fig. 3. Connection wiring diagram of *Dual frequency operation*

two three-phase signals with different frequencies are applied to the two films. The connection for the dual frequency operation is shown in Fig. 3. As explained above, applications of three-phase signals causes traveling potential waves. In this method, due to the different frequencies of the applied signals, the traveling velocities are different for stator and slider, and they are expressed as:

$$v_{sl} = 3pf_{sl} \tag{1}$$

$$v_{st} = spf_{st}, \tag{2}$$

which f_{sl} and f_{st} are the frequencies of applied three-phase signals for slider and stator, respectively. As a result, the operation velocity of the motor becomes:

$$v = v_{sl} - v_{st}, \tag{3}$$

which is the difference of the velocities of the two potential waves.

In this method, effective values of applied signals are not reduced as in *modulated single frequency operation* and therefore the thrust force performance is as good as single frequency operation. Also, the applied frequencies can be chosen arbitrary, since their difference alone has a significant meaning in the motor operation. So, if the frequencies are set high enough, the signals are easily amplified by small-sized transformers and the performance instability at a low speed does not occur.

In single frequency operation and its variant (modulated single frequency operation), the frequency of the driving signals virtually ranges from DC to 1 kHz, which can be easily generated and controlled using a micro controller or a DSP chip. In dual frequency operation, however, two sets of high-frequency sin waves need to be synthesized and they should be precisely controlled in terms of frequency difference or phase difference, to obtain a good motor performance. If the frequency becomes high enough to use small transformers, synthesizing such controlled waves needs some techniques; the previous developments utilized either PLL or DDS. Yamamoto et al. [15] developed an analog driver circuit using PLL. The driver facilitate thrust force control but the analog implementation is less flexible in its configurations and therefore cannot cover a wide range of the motor operation.

Instead of the analog based signal generator, Nishijima et al. [16] and Zhang et al. [17] used Direct Digital Synthesizer (DDS) to generate and control two three-phase signals. The DDS is a digital based generator, so generated signals have high stabilities in a wide range of frequencies. For the motor operation, six DDSs are required. Plus, to synchronize the six DDSs, a micro controller is required. In the previous developments, a micro controller commands initial phase and frequency to six DDSs. But since the communication speed between the DDSs and the controller is limited, the bandwidth of the motor operation is also limited.

III. New driver circuit using FPGA

For dual frequency operation, digital controller using DDS seems quite effective. However, as mentioned in the previous section, the use of commercial DDS chips limit the performance of the motor operation. To overcome this problem, a Field Programmable Gate Array (FPGA) is utilized in this research. Compared with DDS, FPGA is superior in programmability and sampling frequency. In FPGA, DDS circuits and a micro controller to command to the DDSs can be integrated on the same FPGA. Therefore, the communication problem between DDS chips and a micro controller can be easily solved. In addition, the whole system become more simple compared with conventional systems.

A. schematic diagram of the proposed system

Fig. 4 shows the overview of the proposed system. The FPGA board accepts an external control signal via analog/digital ports or serial data connection. This external control signal commands the motor displacement. Based on the external command, the motor driving signals are synthesized. By integrating eq. (5), the following equation is obtained:

$$x = \frac{3p}{2\pi}\phi_{sl} - \frac{3p}{2\pi}\phi_{st} + x_0, \tag{4}$$

where x is the motor displacement and ϕ_{sl} and ϕ_{st} are representative phases of the slider and stator signals respectively. The four DDS modules in the FPGA generates four signals to satisfy the above condition. The resultant signals are:

$$\begin{aligned}
\mathbf{v} &= [\hat{v}_1, \hat{v}_2, \hat{v}_3, \hat{v}_4] \\
&= [\sin(2\pi ft + \phi_{sl}), \sin(2\pi ft + \phi_{sl} - 2\pi/3), \\
&\quad \sin(2\pi ft + \phi_{st}), \sin(2\pi ft + \phi_{st} - 2\pi/3)] \\
&= [\sin(2\pi ft + \phi_{st} + 2\pi x/3p), \\
&\quad \sin(2\pi ft + \phi_{st} - 2\pi/3 + 2\pi x/3p), \\
&\quad \sin(2\pi ft + \phi_{st}), \\
&\quad \sin(2\pi ft + \phi_{st} - 2\pi/3)]
\end{aligned}$$

In the present implementation, the frequency f is manually chosen using the rotary encoder switch. The on-board liquid crystal display (LCD) shows the frequency.

The above signals are output from FPGA through Digital-to-Analog Converters (*DAC*). To save resources of FPGA, only two-third of two three-phase signals ($\sin\omega t$ and $\sin(\omega t - 2/3\pi)$) are generated as shown in the above equation, so the third signals ($\sin(\omega t + 2/3\pi)$) are calculated using op-amps. Then two three-phase signals are amplified by power amplifiers and transformers. Due to the resonance phenomena

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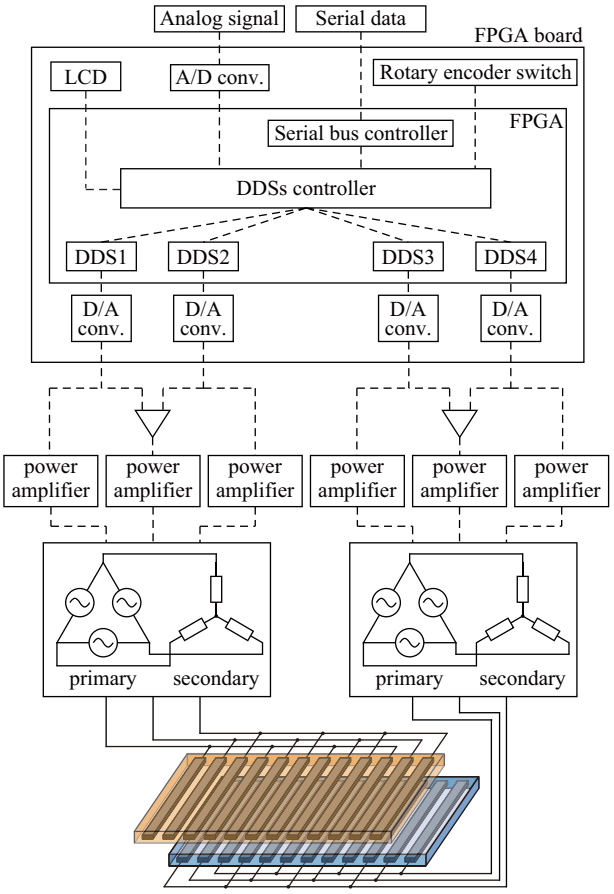


Fig. 4. Electrostatic motor driver circuit using FPGA and transformers

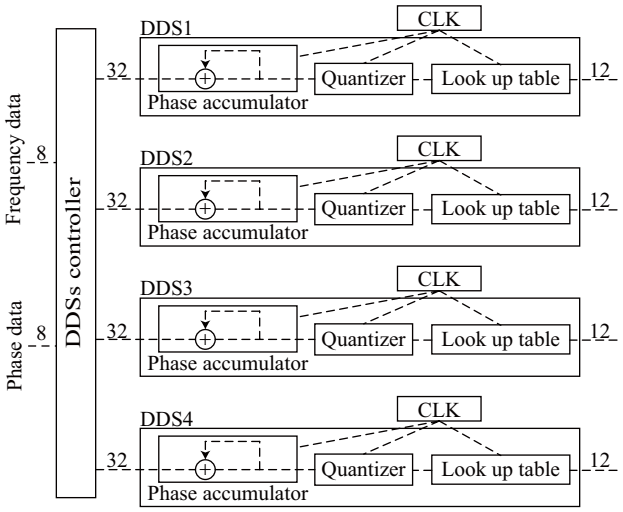


Fig. 5. Implemented modules on FPGA



Fig. 6. Controller board with FPGA

between transformers and the electrostatic motor, reactive currents can be reduced. Transformers are connected each other by δ-Y transformation. In general, the gain of transformers is proportional to their turn ratio, but in case of δ-Y transformation, the gain is equal to the turn ratio times the square root of 3. So δ-Y transformation facilitate the higher amplification gain.

Fig. 5 shows the implemented controller module and DDSs on the FPGA. The numbers shown on signal lines in the Fig. 5 represent the bit length of each signal. The external command signal is collected at regular intervals of the clock pulse. The *Phase accumulator* decides an increment value of an accumulated data using the input signal, then the accumulated data is sent to a Look Up Table (LUT) through a quantizer. In LUT, one cycle's data of the sine wave are stored. Using the accumulated data as a memory address of the LUT, the LUT outputs the sine wave data. Frequencies or phases of generated sine waves are decided and changed by the increment value of the accumulated data from the phase accumulator.

B. FPGA board

Fig. 6 shows the developed motor controller board using the FPGA, which is a combination of XILINX Spartan3 FPGA board and a lab-made Analog-to-Digital Converter (ADC) module. The Spartan3 FPGA board is equipped with RS232C connectors, a dual-channel Analog-to-Digital Converter (ADC), a Quad-channel DAC, and a Static Random Access Memory (SRAM). Table I shows the major specification of the FPGA chip used in Spartan3, including the number of logic cells, Configurable Logic Blocks (CLBs), LUTs, Flip Flops and 18 bit by 18 bit multipliers.

TABLE I
SPECIFICATION OF FPGA

Device	XC3S700AN
Number of Logic Cells	13248
Number of CLBs	1472
Number of LUTs	11776
Number of FFs	11776
Number of MULT18×18s	20

On the FPGA board, one clock bus is shared by three on-board components; the ADC, a programmable gain amplifier for ADC and DAC as shown in Fig. 7. The ADC and the DAC accept up to 50 MHz clock but the programmable amplifier accept up to only 10 MHz, so maximum clock is restricted to 10 MHz if those components are operated at the same time. For the DAC, 10 MHz clock is not enough to

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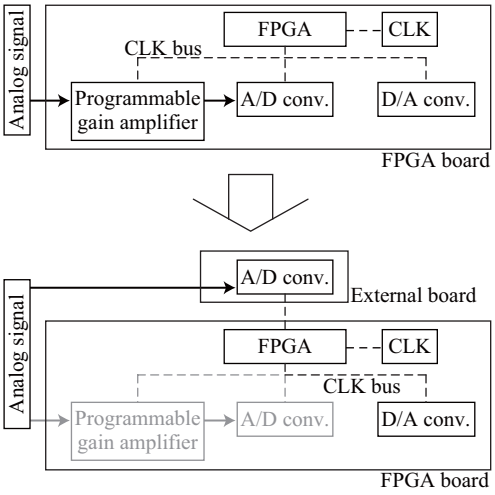


Fig. 7. Controller board with FPGA



Fig. 8. Power amplifiers board

generate quad-channel sine waves with enough accuracy for the motor. Therefore an external ADC board was developed. In our implementation, the external ADC is used instead of the onboard ADC. The programmable amplifier and the ADC on the FPGA board are shut down so that the DAC can operate at 50 MHz.

C. Amplification parts

Fig. 8 shows power amplifier board which contains op-amp circuit to calculate $\sin(\omega t + 2/3\pi)$ from $\sin\omega t$ and $\sin(\omega t - 2/3\pi)$, as well as power op-amps (LM3886, National Semiconductors) to amplify the voltages. One board contains a circuit for one set of three-phase signal, and therefore, two boards are required to operate one electrostatic motor by *Dual frequency operation*. At this board, three-phase signals are amplified up to 70 V_{p-p}. The maximum output current is limited to 4 A.

Fig. 9 shows a transformer board. One board contains three transformers used for one set of three-phase signals. Thus, as same as the amplifier board, two boards are utilized in this work. On this board, primary sides of transformers are connected as delta shape and secondary sides are connected as Y shape. The turn ratio is 1:40 with turn numbers of 21



Fig. 9. Transformers board

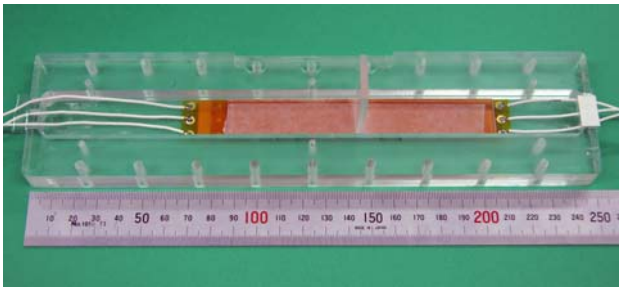


Fig. 10. The electrostatic motor film

and 840 for primary and secondary. The total gain of each transformer is about 70 due to the δ-Y transformation. The transformer cores are PC44 ferrite from TDK with a form factor of PQ5050.

IV. Performance Evaluations

In this section, the proposed system is evaluated using one electrostatic motor and the DSP system.

Fig. 10 shows the electrostatic motor used in the evaluation. These films were fabricated using a flexible printed circuit technology. Electrodes on the film are covered by a polyimide film with a thickness of 15 μm, a length of 135 mm and a width of 20 mm. The pitch of three-phase electrodes are 80 μm.

In this evaluation, the DSP system generates displacement command which is fed to the FPGA controller board through RS232C serial communication. The baud rate of the RS232C communication was set to 115200 bps. The command was in 8-bit format; values from 0 to 255 corresponds to the phase difference of DDSs from 0 to 2π. The relationship between the motor displacement x and the phase difference $\delta\phi$ is $x = 3p\delta\phi/2\pi$. Therefore, the displacement command resolution is $3p/256$ in this configuration, which is almost 1 micron.

Fig. 11 shows evaluation results. As shown in Fig. 11(a) and (b), the motor displacement command was changed continuously by the serial communication. The displacement was measured using a laser displacement sensor (Keyence, LC-2440). Fig. 11(c) shows the measured displacement. Fig. 11 shows that the motor displacement successfully followed the external control command given by the DSP.

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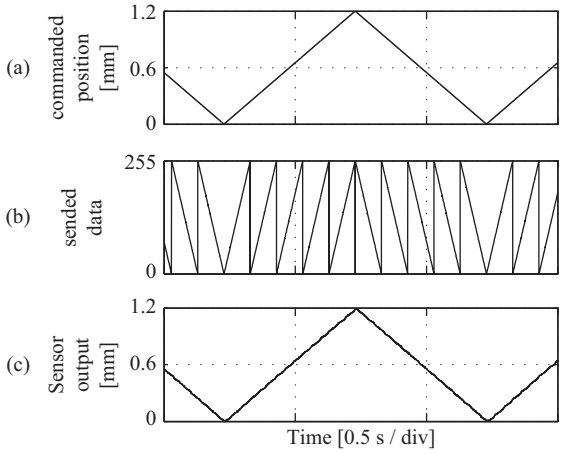


Fig. 11. The electrostatic motor film

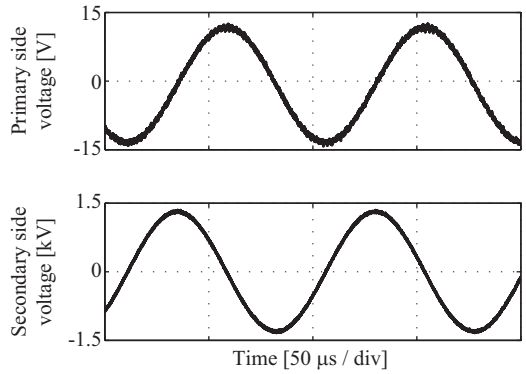


Fig. 12. The electrostatic motor film

One phase of the applied signals to the motor during the operation is shown in Fig. 12. The amplitude of the primary signal, which was the output of the amplifier board, is about 26 V_{p-p}. The secondary side voltage, which is the voltage applied to the motor, was about 1.8 kV_{p-p}, so signal amplification ratio at the transformer was about 69 times.

V. Conclusions

This paper described new electrostatic motor drive circuit using FPGA. To operate the electrostatic motor, both signal synthesizers and the controller are implemented on the same FPGA instead of commercial DDS chips or micro controllers. Implementing the synthesizer on one FPGA facilitate complicate motor controls. Each frequencies or phases can be changed individually using serial communications, analog signals and the rotary switch by hand. The proposed system consists of the FPGA board, the power amplifiers board and the transformers board. In order to save resources of the FPGA, operation amplifiers are implemented to generate the third signal of the three-phase signal on the power amplifier board. The performance was evaluated using the DSP system. It was verified that the driver circuit controlled the electrostatic motor with input data sent by serial communications.

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References

[1] M. Zickar, M. Mita, M. Ataka, and H. Fujita, "Low cross talk design and simple fabrication process of electrostatic vertical comb-drive actuators for positioning application," *IEEJ Transactions on Electrical and Electronic Engineering*, vol. 2, no. 3, pp. 289–294, 2007.

[2] J. D. Grade, K. Y. Yasumura, and H. Jerman, "Micromachined actuators with braking mechanisms," *Sensors and Actuators A: Physical*, vol. 122, no. 1, pp. 1 – 8, 2005.

[3] T. Akiyama, D. Collard, and H. Fujita, "Scratch drive actuator with mechanical links for self-assembly of three-dimensional mems," *Journal of Microelectromechanical Systems*, vol. 6, no. 1, pp. 10 – 17, 1997.

[4] M. Erismis, H. Neves, R. Puers, and C. Van Hoof, "A low-voltage large-displacement large-force inchworm actuator," *Microelectromechanical Systems, Journal of*, vol. 17, no. 6, pp. 1294–1301, Dec. 2008.

[5] L. Frechette, S. Nagle, R. Ghodssi, S. Umans, M. Schmidt, and J. Lang, "An electrostatic induction micromotor supported on gas-lubricated bearings," Interlaken, Switzerland, 2001, pp. 290 – 293.

[6] N. Ghalichechian, A. Modafe, J. H. Lang, and R. Ghodssi, "Dynamic characterization of a linear electrostatic micromotor supported on microball bearings," *Sensors and Actuators A: Physical*, vol. 136, no. 2, pp. 496 – 503, 2007.

[7] M. Beyaz, M. McCarthy, N. Ghalichechian, and R. Ghodssi, "Closed-loop control of a long-range micropositioner using integratedphotodiode sensors," *Sensors and Actuators: A Physical*, vol. 151, no. 2, pp. 187 – 94, 2009.

[8] Y.-C. Tai and R. S. Muller, "Ic-processed electrostatic synchronous micromotors," *Sensors and Actuators*, vol. 20, no. 1-2, pp. 49 – 55, 1989.

[9] S. Nagle, C. Livermore, L. Frechette, R. Ghodssi, and J. Lang, "An electric induction micromotor," *Microelectromechanical Systems, Journal of*, vol. 14, no. 5, pp. 1127–1143, Oct. 2005.

[10] T. Niino, T. Higuchi, and S. Egawa, "Dual excitation multiphase electrostatic drive," *IAS '95. Conference Record of the 1995 IEEE Industry Applications Conference. Thirtieth IAS Annual Meeting (Cat. No.95CH35862)*, vol. vol.2, pp. 1318 – 25, 1995.

[11] T. Niino, T. Higuchi, S. Egawa, and N. Nishiguchi, "Dual excitation electrostatic stepping motor," *Electrical Engineering in Japan*, vol. 119, no. 3, pp. 94 – 103, 1997.

[12] N. Yamashita, Z. G. Zhang, A. Yamamoto, M. Gondo, and T. Higuchi, "Voltage-induction type electrostatic film motor driven by two- to four-phase ac voltage and electrostatic induction," *Sensors and Actuators, A: Physical*, vol. 140, no. 2, pp. 239 – 250, 2007.

[13] R. Gassert, A. Yamamoto, D. Chapuis, L. Dovat, H. Bleuler, and E. Burdet, "Actuation methods for applications in mr environments," *Concepts in Magnetic Resonance, Part B, Magnetic Resonance Engineering*, vol. 29B, no. 4, pp. 191 – 209, 2006/10/.

[14] M. Rajendra, A. Yamamoto, T. Oda, H. Kataoka, H. Yokota, R. Himeno, and T. Higuchi, "Motion generation in mr environment using electrostatic film motor for motion-triggered cine-mri," *Mechatronics, IEEE/ASME Transactions on*, vol. 13, no. 3, pp. 278 –285, 6 2008.

[15] Y. Akio, N. Toshiki, and H. Toshiro, "Development of a controller for a high-power electrostatic motor," *Transactions of the Institute of Electrical Engineers of Japan. D*, vol. 119-D, no. 8/9, pp. pp. 1073–1079, 1999, in Japanese.

[16] T. Nishijima, A. Yamamoto, and T. Higuchi, "An electrostatic motor drive unit with an automatic resonant frequency tuning function," pp. pp. 1151–1152, 12 2008, in Japanese.

[17] Z. Zu Guang, M. Kenji, Y. Akio, and H. Toshiro, "Development of a three-phase high-voltage driving system for thin and flexible film motors," 2010.

[18] T. Niino, A. Yamamoto, and T. Higuchi, "Operation of a dual excitation multiphase electrostatic drive by amplitude-modulated ac voltage," *Elect. Eng. Jpn.*, vol. 131, no. 4, pp. 78–84, 2000.

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